A human dietary risk assessment associated with glycoalkaloid responses of potato to Colorado potato beetle defoliation.

A quantitative human dietary risk assessment was conducted using the glycoalkaloid concentrations measured from tubers of plants defoliated by Colorado potato beetles and undefoliated (control). There was a significantly greater production of glycoalkaloids for defoliated plants compared to control plants for both skin and inner tissue of tubers. The dietary risk posed to different human subgroups associated with the consumption of potatoes was estimated for the 50th, 95th, and 99.9th percentile US national consumption values. Exposures were compared to a toxic threshold of 1.0mg/kg body weight. Defoliation by Colorado potato beetles increased dietary risk by approximately 48%. Glycoalkaloid concentrations within the inner tissue of tubers, including undefoliated controls, exceeded the toxic threshold for all human subgroups at less than the 99.9th percentile of exposure, but not the 95th percentile.